NAME: _____ EXAMINATION NO.: _____

NGULUDI CLUSTER EXAMINATIONS BOARD

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

MATHEMATICS

Subject Number: M131/II

Friday, 27 May

Time Allowed: 2 h 30 mins
8:00 – 10:30 am

PAPER II

(100 marks)

Instructions:

1. **This paper contains 12 printed pages. Please check.**
2. Answer **all** the questions in Section **A** and any **four** questions in Section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided on the question paper.
5. Scientific calculators may be used.
6. **All work must be clearly shown.**
7. Write your **Name** and **Examination Number** at the top of the question paper in the spaces provided.
8. In the table provided on this page, **tick** against the question number you have answered.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

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Section A (60 marks)

Answer all the six questions in this section.

1. **a.** Without using a calculator simplify $\frac{3\sqrt{3}}{7+\sqrt{3}}$ giving the answer with a rational denominator. (*3marks)*

b. Express $\frac{7y}{y^2-49}+\frac{2}{7-y}$ as a single fraction in the lowest form. (*5marks)*

2. **a**. Make y subject of the equation $w=\frac{3yz}{y-3x}$. *(5marks)*

b. $2\log(x+2)+\log 40 = 1+2\log 8$. *(6marks)*

3. a. Find the equation in the form of $y = mx + c$ of a straight line AM which passes through the points (2, 3) and (4, 7). *(5marks)*

3. **b**. Given a polynomial $ax^3 + 3x^2 - 11x + b = 0$ is identical to the polynomial whose roots are 2,-3, and -1/2, calculate the value of a. (*4marks)*

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4. **a**. Two similar solids have volumes of $125cm^3$ and $64cm^3$ respectively. The height of the larger solid is 10cm. what is the height of the other? (*5marks)*

4. b. the length of a cuboid is 18cm, its width is 15cm and is 12cm high.

i. Calculate the length of its longest diagonal.

ii. Calculate the angle between this diagonal and the horizontal. *(7marks*)

5. a. using a ruler and a pair of compasses only, construct in the same diagram:

i. A circle Centre O of radius 4cm.

ii. Two tangents to the circle at S and at R from an external point Q which is 10cm away from the Centre O.

iii. Measure and state the value of angle SQR. (*7marks)*

b. Given that X, Y, Z are matrices such that X=$\begin{bmatrix} 5 & 4 \\ 6 & 6 \end{bmatrix}$, y=$\begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ and 3Z=X-Y, calculate matrix Z. (*4marks)*

6. a. A triangular field has two sides 50m and 60m long and the angle between the sides is 96°. How long is the third side? *(5marks)*

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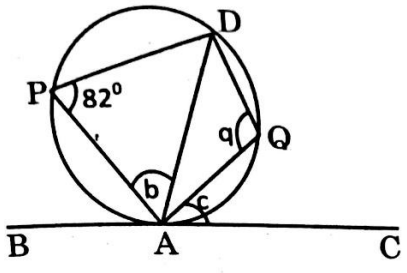
b. In the figure below PD // AQ and AQ =QD and Angle APD is 82°. Calculate the value of angle PAD. (*6marks)*

(b)



SECTION B (40 Marks)

Answer any four questions from this section.

7. The number of students present in a class over a two week period were as follows: 28, 24, 25, 28, 23, 28, 27, 25, 28, 28. Using a formula, calculate the standard deviation for the data. (*10marks)*

8. At Kondanani school, 3m students take chemistry, 65 take physics, 62 take geography, 40 take chemistry and physics, 35 geography and chemistry, 31 take geography and physics, m take all the three subjects and 10 students take neither of them.

i. Draw a Venn diagram representing this information. (*6marks)*

ii. From the Venn diagram, calculate the value of m if there were 125 students at the school. *(4marks*)

9. Yankho has 6 blue marbles, 7 green marbles and 5 white marbles. They are then fully mixed. Two marbles are taken one at a time without replacement.

i. Draw a tree diagram to represent the information. (*4marks)*

ii. Use the tree diagram to calculate the probability that;

a. At least a green marble is picked. (*3marks)*

b. Marbles of the same color are picked. (*3marks)*

10. The table below shows some values of x and y for the equation $y = 2x^2 + x - 2$

x	-3	-2	-1	0	1	2
y	13	4		-2		8

i. Complete the table. (*2marks)*

ii. Use the scale of 2cm to represent 1 unit on the horizontal axis and 2cm m to represent 5units on the vertical axis, draw a graph of $y = 2x^2 + x - 2$ (*5marks)*

iii. Use the graph to solve the simultaneous equation;

iv. $y = 2x^2 + x - 2$.

$y = 2x^2 + 2x - 5$ *(3 mark)*

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11. Solve the simultaneous equations:

$x^2 = y + 5$

$\frac{1}{2}y - x = 1$ *(10marks)*

12. Mr. Nyoni weaves baskets and mats. A mat takes 2 hours of weaving and 5hours of joining. A basket takes 3hours of weaving and 2 hours of joining. He has 60 hours of weaving and 72 hours of joining.

a. Taking x to represent the number of mat and y the number of basket, write down two inequalities in x and y in addition to $y \geq 0$ and $x \geq 0$. *(2marks*)

b. Using a scale of 2cm to represent 5 units on both axes, draw graphs to show the region represented by the four inequalities by shading the unwanted region. (*6marks)*

c. Using the graphs, find the number of mats and baskets Mr. Nyoni can make in full capacity. *(2marks*)

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